FRANCISCO SEABRA SANTOS *BSc Computer Engineering*

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PROFILE

Final-year Computer Engineering student at the University of Coimbra, combining robust software foundations with Artificial Intelligence. Bringing hands-on experience in architecting full-stack systems, deploying AI-driven solutions, and researching Evolutionary Algorithms. Passionate about building scalable, data-driven architectures that leverage machine learning to solve complex problems, and eager to pursue a Master's in Computer Science specializing in AI.

EXPERIENCE

2025 – Present

Importrust, *Freelance AI Engineer & Developer*

- Engineered robust web-scraping solutions to efficiently gather, process, and analyze business-critical data.
- Developed and deployed AI-based tools and custom automation systems to optimize operational efficiency and streamline the company's internal processes.

2024 – Present

jeKnowledge, *Junior Developer*

- Managed technical teams and oversaw full-stack software development lifecycles from conception to delivery.
- Collaborated in *agile* environments to build and deploy web applications focused on sustainability and gamification (see *Projects* section for details).

2024 – 2025

Department of Informatics Engineering Student Union, *Communication Collaborator*

- Managed communication strategies and social media campaigns for major university tech events, including the "*Now What?*" job fair and the "*Coimbra Blockchain*" conference.
- Tailored technical messaging for diverse audiences and corporate partners, enhancing event reach and engagement.

2024

Neuraspace & University of Coimbra, *Undergraduate Research Assistant*

- Selected for a 6-month research grant in Evolutionary Computation during the first year of my bachelor's degree.
- Applied Genetic Algorithms to optimize the routing of a space debris collection prototype, solving a complex aerospace challenge proposed by Neuraspace.
- Conducted applied research under the supervision of Prof. Nuno Lourenço at the Department of Informatics Engineering.

EDUCATION

09/2025 – 01/2026

AI: Financial Markets and Blockchain, *University of Coimbra*
(Average Grade: 18.40)

2023 – 2026
(Expected)

BSc in Computer Engineering, *University of Coimbra*
(Current Average Grade: 15.80)

Relevant Coursework: Feature Engineering for Machine Learning (**17/20**), Discrete Structures (**18/20**), Software Engineering (**17/20**), Mathematical Analysis III (**17/20**), Linear Algebra (**17/20**), Algorithms & Data Structures (**16/20**), Statistics (**16/20**).

2020 – 2023

Science and Technology, *Colégio de São Teotónio*
(Average Grade: 18.85)

TECHNICAL SKILLS

— **Programming Languages:** Python, C, Java, SQL, JavaScript, TypeScript, Assembly

— **AI, Machine Learning & Data Science:** PyTorch, scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Seaborn

— **Web Development:** React, Django, Tailwind CSS, SCSS, shadcn/ui

— **Tools & Databases:** Git, Docker, PostgreSQL, Linux, Jupyter Notebooks, Selenium, Agile, Scrum

RELEVANT PROJECTS

VozSaúde Telehealth System, *AI Engineer & Developer (Social Innovation — "Arrisca C" Competition)*

- Architected an end-to-end telecommunication platform enabling elderly patients to communicate with doctors via automated voice-to-email and email-to-SMS pipelines.
- Engineered a Flask backend to orchestrate asynchronous AI processing (Whisper & LLaMA) seamlessly with Twilio telecom webhooks and custom SMTP/IMAP communication pipelines.
- Currently in the semi-finals of the national entrepreneurship "*Arrisca C*" competition for leveraging AI to solve healthcare accessibility challenges.
- Technologies: Python, Flask, Groq API (Whisper, LLaMA), Twilio, SMTP/IMAP.

Human Activity Recognition System, *Data Scientist & ML Engineer (Academic Project)*

- Developed a machine learning pipeline to classify human activities using time-series data extracted from wearable body sensors.
- Applied advanced feature engineering techniques and classification algorithms to accurately identify and categorize different types of physical movement.
- Technologies: Python, PyTorch, scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn.

APLM, *Full-Stack Developer (jeKnowledge)*

- Engineered a gamified web platform designed to incentivize marine litter collection on beaches to drive measurable environmental impact.
- Integrated a computer vision system to validate user-uploaded photographs of collected trash, automatically awarding points to maximize user engagement.
- Technologies: React, TypeScript, SCSS, Django, Python.

.works Corporate Fair, *Project Manager (jeKnowledge)*

- Led a cross-functional technical team to develop and launch the official website for a major university corporate and job fair.
- Managed the full project lifecycle, overseeing *agile* workflows, team communication, and ensuring a successful, on-time deployment.
- Technologies: React, TypeScript, Tailwind, shadcn/ui, Django, Python, Scrum, Agile, Git.

LANGUAGES

Portuguese — Native

Cambridge Certificate in Advanced English (CAE) — C1
Writing: C2 | Reading: C1 | Speaking: C1 | Use of English: C1 | Listening: C1

AWARDS & INTERESTS

Classical Piano (9 Years), *Colégio de São Teotónio*
Winner of international piano competitions, both solo and 4-hands. Demonstrates long-term dedication, focus, and the ability to perform under pressure.

Competitive Canoe Polo (10 Years), *CFC*
Developed strong teamwork, resilience, and time-management skills by balancing high-level competitive sports with rigorous academic studies.